Muhammad Sami Irfan

+1-205-239-9705 | mirfan@crimson.ua.edu |

Tuscaloosa, AL

Research Interests

Autonomous Vehicle Navigation Security, Transportation Digital Twin

Education

Ph.D. in Civil Engineering *In progress*
The University of Alabama Tuscaloosa, AL, USA

Master's in Civil Engineering *December 2023*
The University of Alabama Tuscaloosa, AL, USA
- GPA: 4.00/4.00

Master's in Business Administration *July 2020*
The University of Dhaka Dhaka, Bangladesh
- GPA: 3.53/4.00

Bachelor of Science in Electrical Engineering *November 2017*
Islamic University of Technology Gazipur, Bangladesh
- GPA: 3.85/4.00

Professional Experience

Graduate Research Assistant *September 2021 - Present*
The University of Alabama Tuscaloosa, AL, USA

Hardware Engineer *January 2020 - August 2020*
MetSys Ltd Dhaka, Bangladesh

Publications and Presentations

Peer reviewed journal articles

(J.1) **Muhammad Sami Irfan**, Sagar Dasgupta, and Mizanur Rahman. Toward Transportation Digital Twin Systems for Traffic Safety and Mobility: A review. IEEE Internet of Things Journal, 2024. DOI: 10.1109/JIOT.2024.3395186

(J.2) S. Dasgupta, **Muhammad Sami Irfan**, M. Rahman, and J. Larson. "GIS and Landmark-Aided Autonomous Vehicle Localization for Navigation Continuity in Urban Corridors." Scientific Reports, Springer Nature, 2026, accepted.

Book chapters

(B.1) Sagar Dasgupta, Xishi Zhu, **Muhammad Sami Irfan**, Mizanur Rahman, Jiaqi Gong, and Steven Jones, "AI Machine Vision for Safety and Mobility: An Autonomous Vehicle Perspective," Handbook on Artificial Intelligence in Transport, Edward Elgar Publishing, 2023.(Peer reviewed)

(B.2) Sagar Dasgupta, **Muhammad Sami Irfan**, Mizanur Rahman, and Mashrur Chowdhury. Modeling, Detection and Mitigation of GNSS Spoofing Attack in Ground Transportation Systems. In Data Analytics for Intelligent Transportation Systems. Elsevier, 2nd edition edition.

Peer reviewed conference publications

(P.1) **Muhammad Sami Irfan**, Sagar Dasgupta, Mizanur Rahman, "A Particle Filter based Sensor Fusion Approach for GNSS Spoofing Detection Incorporating Autonomous Ground Vehicle Navigation Constraints," 2025 IEEE/ION Position, Location and Navigation Symposium (PLANS).

(P.2) S. Dasgupta, M. U. Ahmad, **Muhammad Sami Irfan**, and M. Rahman. "Resilient Urban Vehicle Navigation in GNSS-Denied Environments via Tightly Integrated Inertial Propagation, Deep Motion Inference, GIS Route Constraints, and Opportunistic Visual Landmark Corrections." The 39th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2026), C1: Navigation Resilience to Interference and Cyber-Attacks 1 Technical Session, Orlando, Florida, USA, September 14 - 16, 2026.

(P.3) **Muhammad Sami Irfan**, M. U. Ahmad, S. Dasgupta, and M. Rahman. "A Synchronous GNSS Spoofing Attack Model by Exploiting Basic Safety Message Transmissions for Connected and Automated Vehicles." The 39th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2026), F3: Resilient and Secure PNT Against Interference and Emerging Threats 2 Technical Session, Orlando, Florida, USA, September 14 - 16, 2026.

(P.4) M. U. Ahmad, **Muhammad Sami Irfan**, S. Dasgupta, M. Rahman, and T. N. Bandi. "Beyond Legacy L1 C/A: Characterizing Advanced Multi-Frequency Spoofing Against Multi-Frequency GPS Receivers for Autonomous Navigation." The 39th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2026), C1: Navigation Resilience to Interference and Cyber-Attacks 1 Technical Session, Orlando, Florida, USA, September 14 - 16, 2026.

(P.5) **Muhammad Sami Irfan**, S. Dasgupta, and M. Rahman. "A Particle Filter Based Sensor Fusion Approach for GNSS Spoofing Detection Incorporating Autonomous Ground Vehicle Navigation Constraints." International Technical Meeting of the Institute of Navigation (ION ITM 2025), Long Beach, CA, USA, January 27-30, 2025.

(P.6) S. Dasgupta, **Muhammad Sami Irfan**, and M. Rahman. "Enhanced Navigation for Autonomous Vehicles in GNSS-Denied Urban Environments: Integrating GIS with Deep LSTM/Robust Kalman Filter-Based Landmark Navigation System." International Technical Meeting of the Institute of Navigation (ION ITM 2025), Long Beach, CA, USA, January 27-30, 2025.

Conference presentations (Full manuscript peer reviewed)

(C.1) **Muhammad Sami Irfan**, Mizanur Rahman, Abhay D. Lidbe, Jun Liu, Yangming Shi, Sheila R. Black. "Public Perception of Autonomous Vehicles and Different External Human Machine Interface Designs: Findings from a National Survey," Proceeding of 103rd Annual Meeting of the Transportation Research Board, Washington, D.C., January 2024.

(C.2) **Muhammad Sami Irfan**, Mizanur Rahman, Abhay Lidbe, Jun Liu, "Exploring the Perceptions of Students at Educational Institution Towards Different External Human Machine Interface Designs of Autonomous Vehicles," ASCE International Conference on Transportation & Development, Austin, Texas, June 14-17, 2023.

(C.3) **Muhammad Sami Irfan**, Mizanur Rahman, Travis Atkinson, Sagar Dasgupta, Alex Hainen. "Reinforcement Learning based Cyberattack Model for Adaptive Traffic Signal Controller in Connected Transportation Systems," Proceeding of 102nd Annual Meeting of the Transportation Research Board, Washington, D.C., January 2023.

(C.4) Riffat Islam, Olga A. Bredikhina, **Muhammad Sami Irfan**, Khadiza Tul Jannat, and Steven Jones. "Transportation Needs Assessment for Rural Communities: A Case of Pickens County, Alabama." Proceeding of 102nd Annual Meeting of the Transportation Research Board, Washington, D.C., January 2023.

(C.5) M. Aldeen, L. Cheng, S. Dasgupta, **Muhammad Sami Irfan**, M. Rahman, and M. Chowdhury. "Detection of GNSS Spoofing Attacks Using Vision-Language Models." Transportation Research Board Annual Meeting, Washington, D.C., January 11-15, 2026.

(C.6) S. Dasgupta, **Muhammad Sami Irfan**, and M. Rahman. "Transportation Landmark-based Navigation System for Autonomous Ground Vehicles in GNSS-Contested Urban Areas." Transportation Research Board Annual Meeting, Washington, D.C., January 5-9, 2025.

(C.7) S. Noman, T. Atkison, **Muhammad Sami Irfan**, and M. Rahman. "A Predictive Approach for Sybil attack Detection for a Waiting Time-Based Adaptive Traffic Signal Controller." Transportation Research Board Annual Meeting, Washington, D.C., January 7-11, 2024.

(C.8) M. Rahman, S. Dasgupta, and **Muhammad Sami Irfan**. "Cybersecure GNSS-based Navigation for Autonomous Ground Vehicles." 2024 IEEE Computer Society Annual Symposium on VLSI, ISVLSI 2024, Knoxville, TN, USA, July 1-3, 2024, Invited Talk.

(C.9) S. Dasgupta, A. Ahmed, **Muhammad Sami Irfan**, M. Rahman, and T. Bandi. "Unveiling the Stealthy Threat: Analyzing Slow Drift GPS Spoofing Attacks for Autonomous Vehicles in Urban Environments and Enabling the Resilience." TraCR Annual Conference 2024, Greenville, SC, USA, May 6-7, 2024.

(C.10) S. Dasgupta, K. H. Shakib, **Muhammad Sami Irfan**, and M. Rahman. "Experimental Validation of Sensor Fusion-based GNSS Spoofing Attack Detection Framework for Autonomous Vehicles." TraCR Annual Conference 2024, Greenville, SC, USA, May 6-7, 2024.

Newsletter articles

(N.1) **Muhammad Sami Irfan**, Sagar Dasgupta, and Mizanur Rahman, "Cyberattack Resilient Navigation System for Autonomous Vehicles: An In-Vehicle Sensor Fusion-based Approach," Industrial Control Systems Joint Working Group (ICSJWG) June 2022 Quarterly Newsletter.

Magazine articles

(M.1) **Muhammad Sami Irfan**, Proactive Safety for Vulnerable Road Users Leveraging Digital Twin Technology, Public Roads.

TECHNOLOGY DEMONSTRATIONS

USDOT Future of Transportation Summit 2024 *August 2024*
Securing GNSS-based Autonomous Vehicle Navigation Washington, D.C., USA

TraCR Annual Conference 2024 *May 2024*
Securing GNSS-based Autonomous Vehicle Navigation Greenville, SC, USA

HONORS AND AWARDS

Lifesavers Conference Traffic Safety Scholar *2025*
Lifesavers Conference
- Awarded a scholarship to attend the 2025 Lifesavers conference and recognized as a traffic safety scholar.

Best MS Thesis Award *2025*
The Graduate School, The University of Alabama
- Received the best MS thesis award from the Graduate School at the University of Alabama.

Best MS Thesis Award *2025*
CCEE Department, The University of Alabama
- Received the best MS thesis award from the department of Civil, Construction and Environmental Engineering.

Public Roads Student Writing Competition Award *2023*
Federal Highway Administration (FHWA)
- Won the inaugural public road student writing competition hosted by FHWA. Winning article was featured in the prestigious Public Roads Magazine.

ITE Collegiate Traffic Bowl - ALSITE Traffic Bowl Champion *2023*
The Alabama Section of the Institute of Transportation Engineers
- Won the 2023 ALSITE traffic bowl competition against Auburn University and subsequently represented Alabama at the Southern District ITE traffic bowl competition.

LEADERSHIP EXPERIENCE

Vice-President *2024*
Institute of Transportation Engineers Student Chapter at the University of Alabama

Treasurer *2024*
Bangladesh Student Association at the University of Alabama

MENTORING ACTIVITIES

NSF REU site *2023, 2024, 2025*
The University of Alabama
- Mentored four undergraduate students and developed their research skills

JOURNAL AND CONFERENCE REVIEW SERVICE

Transportation Research Record
IEEE Transactions on Intelligent Vehicles
IEEE Internet of Things
ACM Journal on Autonomous Transportation Systems
Transportation Research Board Annual Meeting
IEEE Vehicular Technology Magazine
Computer Communications
IEEE Open Journal of Intelligent Transportation Systems
IEEE Open Journal of Vehicular Technology

PROFESSIONAL MEMBERSHIPS

American Society of Civil Engineers (ASCE), Student member *Present*
Institute of Transportation Engineers (ITE), Student member *Present*